

System Design (Telusko) - Day 11

One-Shot 5h course | Video 2:30-2:45

Message Queues & Async Processing

Decouple services and smooth spikes with queues.

Key concepts

- Queues (Kafka, RabbitMQ, SQS) buffer work between producers and consumers.
- Async processing improves responsiveness and resilience.
- Pub/sub fans out events to many consumers.
- At-least-once vs exactly-once delivery; idempotency.

Key example

Producer -> queue -> consumers; if a consumer is slow, the queue absorbs the load.

Today's tasks (1.5h)

- ☐ Learn producer/consumer
- ☐ Design an email-send queue
- ☐ Note idempotency

Resources

Video: youtu.be/Vnm-ycSfJx4 (start at 2:30-2:45)

Docs: docs.telusko.com/docs/system-design

(Full clickable links are in the app: Goals > this day > Study block.)